

BSCHE Graduate Written Examination – Fall 2025

Written examination conducted with graduating seniors in Chemical Engineering

Name: Kendall Johnson

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

In the next five to ten years, I would like to be in management, and have a team that focuses on sustainability in diagnostic processes. I want my processes to be both sustainable financially and in material. While recycling diagnostics equipment is a biohazard, I would like to find alternatives to the current plastics being used.

I see all of my student loans paid off, I am pursuing either my masters in chemical engineering, or my MBA. I am looking for companies that support masters programs, and have programs that match my student loan contributions to my retirement accounts.

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Yes, I have looked into getting my PE certification, however I do not currently think this is something I will be pursuing as it is not related to my current fields of interest. If I go to work in power plants, I will revisit this option. I will likely take at least my FE exam.

I would like to go to graduate school in my future, and I will be looking for companies that support this goal.

I have applied to at least 40 jobs at this point in time, gone through two different interview processes, have completed a series of different online assessments as part of the interview process (ranging from one to three hours per assessment), however I do not currently have a job lined up.

Does your prospective employer sponsor master's degree or MBA? Under which terms?

Both of either my masters degree and MBA would be sponsored by Abbott, I would have to work for a minimum amount of time (either one or two years, I do not remember the exact time frame).

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

There have been major deregulatory actions taken with the Environmental Protection Agency, which has and will continue to have major ethical ramifications. Regardless of political stance and belief, it should be agreed upon that waste generation should be avoided within industry, and waste treatment should be mandatory for any processes which generate waste.

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

A lot of the abuse of AI programs that has been occurring during my education is a large concern to me. While I believe AI is a great tool, a lot of critical thinking and problem solving skills are no longer being developed.

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

I approach an ethical dilemma by either discussion with management (be it a professor, or a manager in the work force) and doing my best to set up a clear and logical argument supported by data. My systematic methodology is giving my stance the strongest support I can, and if not heard, proceeding to further escalation, which is something that was instilled in me from my upbringing. My father has been a whistle blower in the past, and stands up for safety.

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

I am familiar with AIChE Code of Ethics. The first provision in it states, "Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties," which is reflected throughout the rest of the provisions within the document. Other notable inclusions are being open to having other people review your work and accepting criticism on it, being truthful and honest – which includes being open to your stakeholders, and upholding professionalism – such as but not limited to; not tolerating harassment and supporting the professional development of those under their supervision.

Which courses at UIC have contributed most to your views on professional ethics? How?

201 – introduction to AIChE and SACHE.

301 – teamwork and team building skills, professionalism

381 – SACHE, lab operations at scale and safety mechanisms in place

396 – HAZOP, and SACHE

Question 3**Working in multidisciplinary teams**

Can you name two other types of engineers that you might find yourself working with?

Mechanical and electrical engineers. (Also, nuclear and process engineers).

In what context or application would you be collaborating with them?

In plant design and plant start up. (Nuclear power plant).

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or VBA): choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} [x_{\text{in}}(t) - x(t)] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

To solve problem (b)

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4			x	y	dx	dy	area		
5			1	0.86376029	0.25	0.83044675	0.20761169		sum
6			1.25	0.7971332	0.25	0.75423115	0.18855779		3.15716526
7			1.5	0.7113291	0.25	0.67742468	0.16935617		
8			1.75	0.64352027	0.25	0.63019656	0.15754914		
9			2	0.61687284	0.25	0.6283235	0.15708087		
10			2.25	0.63977415	0.25	0.67649709	0.16912427		
11			2.5	0.71322004	0.25	0.76579637	0.19144909		
12			2.75	0.8183727	0.25	0.85231702	0.21307926		
13			3	0.88626134	0.25	0.86113398	0.21528349		
14			3.25	0.83600661	0.25	0.76727844	0.19181961		
15			3.5	0.69855026	0.25	0.63187936	0.15796984		
16			3.75	0.56520846	0.25	0.51990562	0.12997641		
17			4	0.47460279	0.25	0.44933199	0.112333		
18			4.25	0.42406119	0.25	0.41245464	0.10311366		
19			4.5	0.40084808	0.25	0.39587414	0.09896853		
20			4.75	0.39090019	0.25	0.38540075	0.09635019		
21			5	0.37990131	0.25	0.36900085	0.09225021		
22			5.25	0.35810039	0.25	0.34241412	0.08560353		
23			5.5	0.32672786	0.25	0.31094775	0.07773694		
24			5.75	0.29516764	0.25	0.28356548	0.07089137		
25			6	0.27196333	0.25	0.26649825	0.06662456		
26			6.25	0.26103318	0.25	0.26198494	0.06549624		
27			6.5	0.26293671	0.25	0.26952584	0.06738146		
28			6.75	0.27611498	0.25	0.28623178	0.07155795		
29			7	0.29634858					
30									

The first step was to set up a series of x values from the range described by the integral (1 to 7), I arbitrarily chose a step of 0.25. I then plugged the x values into the equation within the integral to get my y values. I then multiplied the step factor (0.25) by the average of two consecutive y values to get the area underneath the curve. I summed the averages from 1 to 7 in order to get a final value of approx. 3.16. I plugged the integral into my hand calculator and got a value of approx. 3.15, so I have relative confidence in my answer.

Question 5

Engineering Methodology Proficiency: Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?



Any estimation I could make would be a gross oversimplification of the process. I have no real or meaningful data. I could assume the radius is about “one man” but even then I have no idea if that man is 5’5” or 6’4”. I don’t know the flow rate of the water, and while I know the rate at which the wheel turns, I have no idea of the efficiency of the wheel. I could assume the efficiency is somewhere between 70% and 80%, but that’s also a large guess.